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Chronic Fatigue Syndrome in Long COVID: prevalence and impact on health-related quality of life and health status. A cross-sectional study. BioICOPER study

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Abstract

Background

Long COVID (LC) and Chronic Fatigue Syndrome (CFS) share clinical features and are associated with reduced quality of life and functional impairment. However, the additional burden of LC–CFS overlap remains limited. We aimed to assess the prevalence of LC–CFS overlap and its association with health-related quality of life (HRQoL) and health status outcomes, including sex differences.

Methods

Cross-sectional study including 304 participants with LC. CFS was diagnosed using Fukuda criteria. Health status and HRQoL were assessed with Short Form-36 Survey (SF-36), Goldberg's Anxiety and Depression Scale (GADS), Post-COVID19 Functional Scale (PCFS), Spanish Chronic Pain Grade Scale (SCPGS), modified Medical Research Council Dyspnea Scale (mMRC) and Malnutrition Universal Screening Tool (MUST). Group comparisons and adjusted regression models were performed. In addition, a sex-by-LC-CFS overlap interaction term was included in all regression models.

Results

171 subjects (56.3%) met LC-CFS overlap, more frequently in women (63.8% vs 40.2%). CFS overlap group was related with worse outcomes compared with those with LC alone in most of health questionnaires, including SF-36, GADS, SCPGS, PCFS and mMRC. Compared with LC alone, LC-CFS overlap was associated with poorer scores in SF-36 physical and mental health components (PCS: B=-9.73, MCS: B=-5.36), anxiety (B = 2.02), depression (B = 2.83,95), SCPGS (B = 12.56), PCFS A component (OR = 13.98) and B component (OR 11.27), and mMRC (OR = 7.17). No association was found with MUST or sex interaction.

Conclusions

Over half of participants met criteria for LC–CFS overlap, more frequently in women. This subgroup showed worse HRQoL and health status. Identifying overlap in Primary Care may improve risk stratification and support targeted management strategies.

Trial registration:

This study was carried out with data from subjects of BioICOPER project. The study protocol has been published, and it has been registered in ClinicalTrials.gov in April 2023 with the registration number NCT05819840.

BACKGROUND

Long COVID (LC) has become a major global health concern due to its high prevalence (approximately 7%) and its high individual impact on patients' healthcare, health status, quality of life and economy [1–5]. LC is a complex multisystem disorder [6, 7] characterized by more than 203 reported symptoms affecting multiple organ systems, including the cardiovascular [8], nervous [9, 10], endocrine [11, 12], immune [13, 14], reproductive, and gastrointestinal systems [1]. These manifestations frequently translate into limitations in daily activities, impaired quality of life and increased healthcare need [15]. In addition, most individuals with LC are managed in primary care, buts clinicians frequently face uncertainty regarding its clinical course, prognosis, and management [16, 17].

Preceding the emergence of this condition, Chronic Fatigue Syndrome (CFS) was already known as a chronic multisystemic illness characterized by persistent fatigue lasting for at least six months and not substantially alleviated by rest. This fatigue is accompanied by a range of symptoms, including post-exertional malaise (PEM), unrefreshing sleep and musculoskeletal pain [18, 19], quite similar to LC. Moreover, this condition also leads to a significant reduction in quality of life and functionality, impacting autonomy and physical and mental well-being [15, 20].

Several studies have demonstrated a high similarity between LC and CFS, with symptoms such as PEM, cognitive disorders, chronic fatigue, sleep disruption or functional impairment, affecting more frequently to females with an average age of 30–40 [19, 21–25]. In fact, studies have revealed a prevalence of 40–60% for diagnostic overlap between LC and CFS [25–27]. Both entities generate functional limitations with a significative decreased health-related quality of life (HRQoL), performance of activities of daily living and neuropsychologic state. The presence of fatigue, particularly physical fatigue, has been associated with the manifestation of depression and anxiety symptoms. Moreover, it has been observed to predict a diminished quality of life for individuals across both groups [28]. Additionally, comparative studies have demonstrated that patients diagnosed with LC and CFS present similar levels of impairment in all domains of quality of life as measured by validated questionnaires such as the SF-36 questionnaire [20, 29]. This spiking resemblance has created interest in pursuing further research in this area, with a particular focus on the overlap between these two conditions.

On the other hand, there is no consensus in the exact etiology of LC or CFS, although they share several physiopathology mechanisms such as immune abnormalities, mitochondrial disfunction, neuroinflammation [18, 19, 27] and endothelial dysfunction, particularly affecting the microvasculature and associated with low-grade inflammation [30–34]. Furthermore, evidence suggests that both entities are infection-associated conditions, characterized by initial infections that trigger the downstream pathological mechanisms, resulting in chronic multisystem dysfunction. This strengthens the hypothesis of a convergent pathophysiology [24, 25].

A deeper knowledge of the overlapping between LC and CFS as well as the factors related to their health status and HRQoL provides an essential foundation for the formulation of targeted, evidence-informed interventions and for optimizing quality-of-life outcomes among affected populations [24, 35].

Nevertheless, there is insufficient evidence regarding susceptibility factors and the differences in perceived health status between patients who meet overlap criteria and those who do not. This underscores the necessity for further research in this area [25, 26].

Therefore, this study aims to investigate the overlap between LC and CFS diagnosis and to assess whether their coexistence is associated with poorer HRQoL and health status than LC alone, as well as to explore potential sex differences.

METHODS

Study design

This cross-sectional observational study was carried out with data from subjects of BioICOPER project at the Salamanca Primary Care Research Unit (APISAL) and the study protocol has been published [36].

Study population

304 participants were included with a diagnosis of LC in the Primary Care and University Hospital of Salamanca registries, between 09/03/2023 and 10/09/2024. Recruitment was carried out by consecutive sampling as shown in Fig. 1. The diagnosis of LC was established according to the diagnosis criteria of the World Health Organization (WHO) [37]: symptomatology of at least 2 months' duration, appearing within 3 months of SARS-CoV-2 infection with no other apparent cause. These symptoms may be fluctuating or persist after acute infection [37]. Subjects with established cardiovascular disease, terminally ill, unable to travel to the health center, or those with a glomerular filtration rate lower than 30 mL/min/1.73m2 were excluded.

Variables and measurement instruments

To reduce information bias, the variables analysed, examinations and tests were carried out by four trained healthcare professionals, following a standardized protocol [36]. The information was recorded in the REDCAP system and data quality control was performed by an independent researcher.

Sociodemographic variables

The following data of the participants were collected: age, sex and the date on which the acute infection of the disease was diagnosed. This was then employed to calculate the time taken for the disease to progress [36].

Number of components of Metabolic Syndrome

Metabolic Syndrome (MetS) components were defined according to the Joint Scientific Statement of the National Cholesterol Education Program Adult Treatment Panel III (NCEP ATP III) criteria [38]. The five components were: (1) blood pressure ≥ 130/85 mmHg or current antihypertensive treatment; (2) fasting plasma glucose ≥ 100 mg/dL or use of glucose-lowering medication; (3) triglycerides ≥ 150 mg/dL or use of lipid-lowering therapy; (4) HDL cholesterol < 40 mg/dL in men or < 50 mg/dL in women; and (5) waist circumference ≥ 102 cm in men or ≥ 88 cm in women.

Waist circumference (WC) was assessed with a tape measure above the iliac crest at the end of expiration and with the patient standing and unclothed.

Blood pressure (BP) was measured with a validated Omron M10-IT sphygmomanometer (Omron Healthcare, Kyoto, Japan), with the average of the last 2 readings recorded. Measurements were taken on the participant's dominant arm in a sitting position after at least 5 min of rest, using an appropriately sized cuff, determined by measuring the circumference of the upper arm, following the recommendations of the European Society of Hypertension (ESH) [39].

Plasma glucose, triglycerides, total cholesterol and high-density lipoprotein (HDL-c) were obtained through blood samples collected in the morning after fasting overnight, utilizing an automated standardized enzyme assay (Atellica™ Solution device. SIEMENS S.A—Erlangen, Germany—ISO 15189) [40].

Use of glucose-lowering, lipid-lowering, and antihypertensive medications were assessed by direct questioning of participants and verified using the electronic health record.

Persistent symptoms reported and medical history

A questionnaire-guided anamnesis, developed for the BioICOPER project [36], was conducted to collect information on persistent symptoms relevant to CFS and LC diagnosis: fatigue, lack of energy, memory loss, difficulty concentrating, dyspnea, sore throat, muscle pain, joint pain, headache and sleep disturbances. Participants were asked about their medical history, including asthma, chronic obstructive pulmonary disease (COPD), and cancer. An English-language version of this study-specific form is provided as Additional file 1.

Health status questionnaires

The following previously published and validated standardized questionnaires were used to assess HRQoL and health status.

HRQoL was assessed with the Short Form Health Survey (SF-36) questionnaire [41]. It is composed by 36 items regarding characteristics of 8 sections of health status for the last 4 weeks: vitality, physical functioning, physical role functioning, bodily pain, general health perceptions, emotional role functioning, social role functioning and mental health. Each section is scored from 0 to 100 with a result of 100 indicative of optimal health. The 8 section scores are standardized with Z-score for general Spanish

population [41] utilizing Spanish weights published by Vilagut et al [42]. The different sections are unified in two components: Physical Component Summary (PCS) and Mental Component Summary (MCS)[43, 44]. In addition, self-perceived change in health status was evaluated using the Health Transition Score (HTS) of the SF-36 questionnaire.

Dyspnea was evaluated with the modified dyspnea scale (mMRC), graded from 0 to 4, with higher scores reflecting more severe breathlessness [45].

Fatigue was assessed with Modified Fatigue Impact Scale (MFIS) [46]. This questionnaire is composed by 21 items and is divided in 3 dimensions of fatigue: physical (9 items), cognitive (10 items) and psychosocial (2 items). Each item is rated from 0 (never) to 4 (almost always) and the total score ranges from 0 to 84. Higher scores indicate a greater fatigue on daily life. MFIS ≥38 was the cutting point due to it is often used to distinguish people with moderate or severe fatigue from mild symptoms [46, 47].

Sleep Quality was evaluated with Pittsburgh Sleep Quality Index (PSQI) [48] which consists of 10 questions and assesses the quality of patient's sleep during the last month. The total score ranges from 0 to 21 points, with higher scores indicating worse sleep quality [48], where > 5 points was used as a cutting point as it means poor sleep quality.

The assessment of *pain* was conducted employing the Spanish Chronic Pain Grade Scale (SCPG) [49]. It comprises 8 items regarding the presence or absence of pain in the patient, its intensity, and its impact on the performance of activities of daily living, recreational and professional activities over the previous three months. Each item is assigned a score ranging from 0 to 10. The total score ranges from 0 to 40. Scores obtained are directly proportional to the pain experienced.

Depression and *anxiety* were assessed employing the Spanish version of Goldberg's Anxiety and Depression Scale (GADS) [50, 51]. It is composed of two subscales: the first one concerns anxiety (questions 1–9) and the second relates to depression (questions 10–18). Elevated values are congruent with diminished emotional well-being.

Malnutrition was evaluated with the Malnutrition Universal Screening Tool (MUST) [52, 53]. It assesses the weight loss and BMI for the last 3–6 months, and the intake capacity in the last 5 days. Subjects were labeled into two categories: low risk (0 points) and intermediate-high risk (≥ 1 point).

The functional evaluation was conducted utilizing the Post COVID-19 Functional Status Scale (PFSC) [54]. The questionnaire is composed of two sections (a and b). Component A assesses the degree of functional limitations in daily activities due to persistent symptoms, whereas component B evaluates symptom exacerbation following physical or mental exertion. Both were measured from grade 1 to 4 in accordance with the patient's limitations. The higher score, the worse functionality [55]. A cut-off of ≥ 2 points was applied to identify the subjects with poor functional status.

Chronic Fatigue Syndrome definition

Fukuda criteria [56] were employed in order to define CFS, as illustrated in Table 1.

Exertional intolerance was operationalized using two MFIS items addressing the need to reduce physical activity pace and the difficulty in sustaining physical effort over time (Item 6: “I have had to moderate my pace in physical activities”; item 10: “I have trouble maintaining physical effort for long periods of time”). These items were used as indicators of activity-related fatigability, a core functional feature in CFS-like presentations.

Table 1
Diagnostic criteria for Chronic Fatigue Syndrome (CFS)

Criteria	Measurement tool / items	Cut-off / specification
Persistent fatigue > 6 months	MFIS, symptom report of persistent fatigue.	MFIS ≥ 38 or reported symptom of persistent fatigue
Unexplained after clinical evaluation	Clinical evaluation	All participants (it is part of LC definition)
New onset (not lifelong)	Clinical history	All participants with persistent fatigue
Presence of four or more of the following:		
1. Exertional intolerance (proxy for PEM)	MFIS items 6 and 10*	≥ 2 points in both items or 3–4 points in one of them.
2. Unrefreshing sleep	PSQI > 5, symptom report	Presence of one or both
3. Impaired memory or concentration	Symptom report	Presence of one or both
4. Muscle pain	Symptom report	Present
5. Pain in joints	Symptom report	Present
6. New headaches	Symptom report	Present
7. Sore throat	Symptom report	Present
8. Tender cervical or axillary lymph nodes	Not included	Not applicable for LC

*Item 6: “I have had to moderate my pace in physical activities”; item 10: “I have trouble maintaining physical effort for long periods of time”. MFIS: Modified Fatigue Impact Scale; LC: Long Covid; PEM: post-exertional malaise; PSQI: Pittsburgh Sleep Quality Index.

Statistical Analysis

A target sample size of 300 participants was established a priori. With a two-sided alpha level of 0.05, this sample size provided 80% power to detect a small-to-moderate standardized difference (Cohen’s d = 0.32) in HRQoL outcomes between participants with LC-CFS overlap and those with LC alone.

The Kolmogorov–Smirnov test was used to assess the normality of the variables. Student's t-test, Chi-square test, or Mann–Whitney U test was applied, as appropriate, to compare health status questionnaire scores and proportions, first in the overall sample and then according to the presence of overlapping LC and CFS diagnoses, depending on the type and distribution of the variables. All variables were analysed disaggregated by sex. Continuous questionnaire scores (SF-36, SCPGS and GADS) were analysed using multiple linear regression models. Ordinal logistic regression models were initially fitted for mMRC, PCFS and MUST. When the proportional odds assumption was violated (PCFS) or when model instability occurred due to sparse data (MUST), variables were dichotomized and analysed using binary logistic regression. Due to the low prevalence of high malnutrition risk (MUST ≥ 2, n = 11), MUST was dichotomized as moderate-high nutritional risk (≥ 1) versus no risk (0) for logistic regression analyses. All models included LC-CFS overlap as the main independent variable and were adjusted for sex (0 = male, 1 = female), time evolution since acute infection, number of components of MetS, the presence of cancer, COPD, and asthma (all encoded 0 = no, 1 = yes). Effect modification by sex was assessed by including a sex-by-LC-CFS overlap interaction term in all regression models.

The subsequent analysis of the data was conducted utilising the SPSS v30.0.0.0 statistical software package, specifically designed for utilisation on the MacOS operating system (IBM, Armonk, New York: IBM Corp). The statistical significance was set at p value < 0.05.

Use of artificial intelligence tools

During manuscript preparation, ChatGPT (OpenAI) was employed to assist with English-language editing and with the formatting and assembly of manuscript figures based on author-generated outputs. No generative AI was used to create scientific images, generate data, perform analyses, or alter study results. All figures were checked against the original data and statistical outputs by the authors. The authors reviewed and approved all AI-assisted edits and take full responsibility for the final manuscript.

RESULTS

As illustrated in Table 2, the subjects collected in the present study were analyzed on a global scale and according to sex (women and men). A total of 304 participants were included, with a greater proportion of female participants (207 vs 97). Women were younger than men (51.3 ± 11.5 vs 55.7 ± 12.3 years), while time since acute infection did not differ. Cardiometabolic risk factors were more prevalent among men, including hypertension, diabetes, and greater waist circumference. Men showed a higher use of antihypertensive, lipid-lowering, and hypoglycemic medications than women.

Table 2
General characteristics of the participant subjects, symptoms reported and health status questionnaires.

General characteristics	Global	Men	Women	*p* value
	Mean ± SD or n (%)	Mean ± SD or n (%)	Mean ± SD or n (%)	
N = 304	**N = 97**	**N = 207**		
Age (years), mean ± SD	52.71 ± 11.94	55.70 ± 12.28	51.32 ± 11.54	0.001
Time period since acute infection (months), mean ± SD	38.66 ± 9.58	38.74 ± 9.41	38.50 ± 9.96	0.990
CVF				
HAH (mmHg) n (%)	109 (35.9)	52 (53.6)	57 (27.5)	< 0.001
Dislipemia (mg/dl) n (%)	172 (56.6)	67 (69.1)	105 (50.7)	0.003
Total cholesterol, (mg/dl), mean ± SD	187.45 ± 34.30	182.11 ± 32.94	189.95 ± 34.71	0.029
HDL cholesterol, (mg/dl), mean ± SD	56.84 ± 13.53	48.78 ± 10.86	60.764 ± 13.02	< 0.001
Triglycerides, (mg/dl), mean ± SD	102.23 ± 50.81	117.47 ± 54.39	95.09 ± 47.52	< 0.001
FPG, (mg/dl), mean ± SD	87.88 ± 17.67	94.37 ± 19.78	84.84 ± 15.74	< 0.001
Diabetes mellitus, n (%)	37 (12.2)	22 (22.7)	15 (7.3)	< 0.001
Waist circunference (cm), mean ± SD	93.92 ± 15.49	104.34 ± 12.52	89.04 ± 14.31	< 0.001
MetS components, mean ± SD	1.54 ± 1.34	2.04 ± 1.38	1.3 ± 1.26	< 0.001
MetS, n (%)	72 (23.7)	39 (40.2)	33 (15.9)	< 0.001
Medical history				
Cancer n (%)	18 (5.9)	9 (9.3)	9 (4.3)	0.090
COPD n (%)	5 (1.6)	3 (3.1)	2 (1.0)	0.174
Ashtma n (%)	47 (15.5)	10 (10.3)	37 (17.9)	0.089
Cardiometabolic medications				

General characteristics	Global	Men	Women	*p* value
	Mean ± SD or n (%)	Mean ± SD or n (%)	Mean ± SD or n (%)	
N = 304	**N = 97**	**N = 207**		
Antihypertensive drugs n (%)	79 (26)	34 (35.1)	45 (21.7)	0.014
Lipid-lowering drugs n (%)	75 (24.7)	40 (41.2)	35 (16.9)	0.01
Hypoglycemic drugs n (%)	32 (10.5)	18 (18.6)	14 (6.8)	0.002

SD: Standard deviation.; CVF: Cardiovascular Risk Factors; HAH: High Arterial Hypertension; HDL: high–density lipoprotein; FPG: fasting plasma glucose; BMI: Body mass index; MetS components: number of Metabolic Syndrome components; COPD: Chronic Obstructive Pulmonary Disease; *p* value: differences between men and women.

Table 3 details reported symptoms and health status questionnaires, where women reported higher frequencies of fatigue (75.4% vs 62.9%), cognitive impairment (65.7% vs 47.4%), musculoskeletal pain, and sleep disturbances. Women also showed worse fatigue (MFIS ≥ 38: 43.8% vs 14.5%), poorer sleep quality (PSQI > 5 60.1% vs 24.5%), higher anxiety and depression scores, and greater functional limitation. On the other hand, men had higher SF-36 component scores (PCS: 42.8 ± 10.8 vs 38.6 ± 10.6 and MCS: 44.4 ± 11.4 vs 40.7 ± 13.5). A similar pattern was observed across the eight SF-36 domains, as detailed in Table S1 (Additional File 2) No differences between sex were observed for Health Transition Score (HTS), MUST or mMRC.

Figure 2 illustrates results of the classifications of the individuals according to fulfilment of criteria for LC and LC-CFS overlap. 171 subjects (56.3%) met the overlap diagnosis criteria, with a higher proportion of women than men.

Table 4

summarizes the comparisons of health outcomes between patients with LC alone and those with LC–CFS overlap. Participants with LC–CFS overlap presented significantly worse scores compared with those with LC alone in most questionnaires, including SF-36, GADS anxiety and depression components, SCPGS and both components of PCFS. Differences between overlapping and non-overlapping diagnosis showed $p < 0.001$ in all cases, except for MUST. In addition, as detailed in Tables S2 and S3 (Additional file 2), similar patterns to the overall results were observed in both women and men.

Variable	**Global**	**Men**	**Women**	***p* value**
Symptoms reported	**Mean ± SD or n (%)**	**Mean ± SD or n (%)**	**Mean ± SD or n (%)**	
	N = 304	**N = 97**	**N = 207**	
Fatigue n (%)	217 (71.4)	61 (62.9)	156 (75.4)	0.025
Memory impairment n (%)	182 (59.9)	46 (47.4)	136 (65.7)	0.002
Concentration impairment n (%)	171 (56.3)	39 (40.2)	132 (63.8)	< 0.001
Joint pain n (%)	169 (55.6)	36 (37.1)	133 (64.3)	< 0.001
Muscle pain n (%)	165 (54.3)	38 (39.2)	127 (61.4)	< 0.001
Headache n (%)	121 (39.8)	29 (29.9)	92 (44.4)	0.016
Sleep disorders n (%)	184 (60.5)	44 (45.4)	140 (67.6)	< 0.001
Sore throat n (%)	63 (20.7)	17 (17.5)	46 (22.2)	0.346
Health questionnaires				
MFIS ≥38, n (%)	177 (58.2)	44 (14.5)	133 (43.8)	0.002
PSQI > 5, n (%)	252 (82.9)	73 (24.5)	179 (60.1)	0.005
SF-36 PCS, mean ± SD	39.94 ± 10.82	42.79 ± 10.83	38.61 ± 10.58	0.001
SF-36 MCS, mean ± SD	41.86 ± 12.98	44.42 ± 11.41	40.67 ± 13.52	0.007
SF-36 HTS, mean ± SD	50.34 ± 25.46	51.84 ± 25.06	49.63 ± 25.67	0.486
GADS anxiety, mean ± SD	3.94 ± 2.28	3.03 ± 3.15	4.37 ± 3.27	< 0.001
GADS depression, mean ± SD	3.63 ± 2.63	2.69 ± 2.65	4.07 ± 2.51	< 0.001

Variable	Global	Men	Women	*p* value
Symptoms reported	**Mean ± SD or n (%)**	**Mean ± SD or n (%)**	**Mean ± SD or n (%)**	
	N = 304	**N = 97**	**N = 207**	
SCPGS, mean ± SD	29.67 ± 17.80	23.60 ± 17.27	32.56 ± 17.36	< 0.001
PCFS(a)≥2 n (%)	210 (69.1)	67 (25.2)	143 (74.8)	< 0.001
PCFS(b)≥2 n (%)	186 (61.2)	48 (49.5)	138 (66.7)	0.004
MUST ≥1 n (%)	29 (9.5)	6 (6.2)	23 (11.1)	0.173
mMRC, mean ± SD	1.22 ± 0.42	1.21 ± 0.41	1.23 ± 0.42	0.382

Table 3. Symptoms reported and health status questionnaires scores.

Values are means and standard deviations for continuous data, and number and proportions for categorical data. SD: Standard deviation; MFIS: Modified Fatigue Impact Scale; PSQI: Pittsburgh Sleep Quality Index; SF-36: Short Form-36 Health Survey; PCS: Physical Health Component Summary; MCS: Mental Health Component Summary; HTS: Health Transition Score; GADS: Goldberg Anxiety and Depression Scale; SCPGQ: Spanish Chronic Pain Grade Questionnaire; PCFS: Post-COVID Functional Status Scale (a): a component; (b): b component; MUST: Malnutrition Universal Screening Tool; mMRC: modified Medical Research Council Scale; *p* value: differences between men and women

MFIS: Modified Fatigue Impact Scale; CFS: Chronic Fatigue Syndrome. *p* value: differences between Men and Women; *$p < 0.05$; **$p < 0.001$ for sex differences.

Figure 3 presents the multiple linear adjusted regression model examining the association between LC–CFS overlap and health questionnaire outcomes. LC–CFS overlap was associated with poorer HRQoL across all SF-36 domains, as well as GADS anxiety and depression components and pain scale SCPGS.

Figure 4 shows the results of the binary and ordinal regression models. LC–CFS overlap was associated with functional limitation (PCFS (a) ≥ 2: OR = 13.98 (4.64 to 42.13) PCFS (b) ≥ 2: OR = 11.27, 95% CI 4.08–31.17) and greater dyspnea severity (mMRC: OR = 7.17, 95% CI 3.06–16.79), whereas no significant association was observed for nutritional risk.

No evidence of effect modification by sex was found in the sex-by-overlap interaction analysis ($p > 0.05$).

Table 4
Comparison of health status questionnaires outcomes between patients with LC and LC-CFS overlap

Health status questionnaires	No ovelap	Overlap	*p* value
	Mean ± SD or n (%)	Mean ± SD or n (%)	
N = 133	**N = 171**		
SF-36 PCS, mean ± SD	46.18 ± 8.50	35.04 ± 9.9	< 0.001
SF-36-MCS, mean ± SD	45.41 ± 11.46	39.08 ± 13.46	< 0.001
SF-36 HTS, mean ± SD	56.87 ± 23.21	45.15 ± 26.03	< 0.001
SF-36 physical functioning, mean ± SD	83.77 ± 18.00	61.54 ± 23.94	< 0.001
SF-36 role physical, mean ± SD	72.88 ± 37.87	30.87 ± 39.87	< 0.001
SF-36 bodily pain, mean ± SD	61.95 ± 22.53	36.11 ± 21.76	< 0.001
SF-36 general health, mean ± SD	55.93 ± 18.86	36.97 ± 17.73	< 0.001
SF-36 vitality, mean ± SD	53.21 ± 20.49	30.99 ± 17.76	< 0.001
SF-36 social functioning, mean ± SD	7700 ± 22.56	54.89 ± 26.85	< 0.001
SF-36 role emotional, mean ± SD	76.67 ± 36.33	53.61 ± 45.80	< 0.001
SF-36 mental health, mean ± SD	68.58 ± 16.53	56.06 ± 20.31	< 0.001
GADS Anxiety, mean ± SD	2.90 ± 2.86	4.75 ± 3.38	< 0.001
GADS Depression, mean ± SD	2.22 ± 2.55	4.73 ± 2.12	< 0.001
SCPGS, mean ± SD	19.62 ± 15.12	37.49 ± 15.71	< 0.001
PCFS (a) ≥2, n (%)	60 (28.6)	150 (71.4)	< 0.001
PCFS (b) ≥2, n (%)	41 (22)	145 (78)	< 0.001
MUST ≥1, n (%)	15 (51.7)	14 (48.3)	0.363
mMRC,) mean ± SD	0.7 ± 0.74	1.34 ± 0.85	< 0.001

Values are means and standard deviations for continuous data, and number and proportions for categorical data. SD: Standard deviation; LC: Long COVID, CFS: Chronic Fatigue Syndrome; SD: Standard deviations; SF-36: Short Form-36 Health Survey; PCS: Physical Component Summary; MCS: Mental Component Summary; HTS: health transition score; SCPGQ: Spanish Chronic Pain Grade Questionnaire; GADS: Goldberg Anxiety and Depression Scale; PCFS: Post-COVID Functional Status Scale (a): a component; (b): b component; MUST: Malnutrition Universal Screening Tool; mMRC: modified Medical Research Council Scale; *p* value: differences between LC and LC-CFS overlap.

LC: Long COVID; CFS: Chronic Fatigue Syndrome; SF-36: Short Form-36 Health Survey; PCS: Physical Component Summary; MCS: Mental Component Summary; HTS: health transition score; GADS: Goldberg Anxiety and Depression Scale; SCPGQ: Spanish Chronic Pain Grade Questionnaire.

DISCUSSION

In this study, 56.3% of patients with LC fulfilled diagnostic criteria for CFS overlap, with a higher prevalence observed in women than men. This subgroup exhibited markedly poorer HRQoL and overall health status compared with patients without overlap, as reflected by consistently lower scores on the SF-36, GADS, SCPGS, PCFS and mMRC questionnaires. MUST outcomes and sex-based differences did not differ between groups.

LC: Long COVID; CFS: Chronic Fatigue Syndrome; PCFS: Post-COVID Functional Status Scale; (a): a component; (b): b component; MUST: Malnutrition Universal Screening Tool; mMRC: modified Medical Research Council Scale.

These results are consistent with previous evidence suggesting that meeting diagnostic criteria for CFS in addition to LC may reflect a more severe and chronic phenotype characterized by worse quality of life [57, 58]. Growing evidence indicates substantial clinical convergence between LC and CFS regarding persistent fatigue, exertional intolerance, cognitive dysfunction, sleep disturbance, and pain [59–61]. Likewise, several studies have reported that a similar proportion of individuals with LC meet established CFS criteria [21, 25, 62].

However, this study extends the literature by indicating that this overlap is not merely diagnostic but translates into significantly worse patient-reported outcomes across multiple independent domains of health. Importantly, the findings suggest that the LC–CFS overlap phenotype is not merely a diagnostic construct but represents a clinically meaningful subgroup characterized by considerably worse multidimensional health outcomes. The consistent differences observed across anxiety, pain, and functional scales suggest that LC–CFS overlap reflects a broader systemic burden rather than isolated symptom amplification. This is consistent with the results reported by Garapaty et al. [57] where individuals with overlapping CFS and LC exhibited a greater burden of cognitive and psychological symptoms, including memory impairment, anxiety, depression, and fatigue.

Health related quality of life

Patients with LC–CFS overlap showed lower scores across all SF-36 domains compared with those with LC alone, indicating a generalized and profound impairment in HRQoL. In addition, LC-CFS overlap was associated with a worse perceived health transition score HTS, indicating a greater subjective deterioration compared with the previous year. Reduced SF-36 scores have been consistently reported in both LC and CFS populations. In the widely cited review by Nalbandian et al. [63], persistent functional limitation and reduced quality of life were highlighted as central features of LC. Similarly, previous

research has demonstrated that CFS is associated with quality-of-life impairments comparable to or even exceeding those observed in other chronic medical conditions [19, 57, 59].

The additive reduction observed in the overlap group may therefore reflect the convergence of two pathophysiological processes with partially shared mechanisms, including autonomic dysregulation, neuroimmune activation, and central sensitization [18, 19, 64]. In this context, the global deterioration across domains supports the hypothesis that LC patients fulfilling CFS criteria represent a clinically distinct and more severely affected phenotype. Consistent with this interpretation, prospective and cohort studies have shown that LC individuals meeting CFS criteria exhibit greater symptom burden, more pronounced autonomic dysfunction, and significantly worse functional status compared to other LC patients [57, 62].

The impairment observed in the mental health and role-emotional domains may reflect the psychosocial consequences of persistent disability and sustained functional restriction. Previous studies in CFS populations have reported that prolonged symptom burden and physical impairment are associated with anxiety, depressive symptoms, and reduced social participation [65, 66]. Nevertheless, bidirectional interactions between physical and psychological factors cannot be excluded, as emotional distress may both result from and contribute to perceived functional limitation.

Functional impairment

From a functional perspective, LC-CFS overlap group obtained higher grades on the PCFS, which reinforces the clinical significance of overlap status. To date, direct specific comparative analyses of functional status between LC patients with and without CFS overlap remain scarce. However, previous research has identified a subset of Long COVID patients fulfilling CFS criteria where worse health status was reported [57, 60, 64]. Given that activity intolerance is a core features of CFS, it is biologically plausible that LC patients meeting these criteria would experience greater participation restriction and daily activity limitation. Moreover, the association with higher PCFS grades but not with nutritional risk suggests that the observed impairment is primarily functional and systemic rather than driven by frailty or malnutrition.

Anxiety and psychological burden

Patients with LC–CFS overlap demonstrated higher scores on the GADs anxiety and depression subscales. These findings are consistent with those reported by Garapaty et al. [57], who observed higher levels of anxiety and depression among individuals with LC–CFS overlap compared with those with LC alone (OR 7.11, 95% CI 4.23–11.95 vs OR 1.84, 95% CI 1.62–2.08). In contrast, a study conducted in an Australian population did not find significant differences in the Hospital Anxiety and Depression Scale (HADS) anxiety subscale between individuals with LC–CFS overlap and those with LC alone [64]. This discrepancy may be explained by the smaller sample size of that study.

Although psychological symptoms should be carefully interpreted as potential primary etiological drivers, they are increasingly recognized as important components of illness burden in chronic post-

infectious syndromes [63]. Elevated anxiety may reflect uncertainty regarding prognosis, prolonged functional impairment, and reduced participation in social and occupational activities. The bidirectional interaction between anxiety, sleep disturbance, and physical symptoms likely amplifies overall disease impact, which underscores the importance of integrated mental health support within Long COVID management.

Pain burden

The findings suggest that LC patients meeting CFS criteria experience a greater pain burden, consistent with a more severe post-viral phenotype. In a large survey study comparing LC, CFS and fibromyalgia, comorbid presentations (LC + CFS and/or fibromyalgia) showed higher pain severity and greater pain-related functional interference compared with LC alone [61]. These observations align with other studies in CFS populations that indicate that co-occurring pain conditions are associated with worse pain metrics and poorer functioning, including lower SF-36 Bodily Pain scores, reinforcing the plausibility of pain amplification within the CFS spectrum [67, 68]. The greater pain burden observed in LC–CFS overlap patients may reflect shared mechanisms of central sensitization, autonomic dysfunction, and neuroinflammation previously described in CFS populations [19, 23, 59].

Sex differences and effect modification

Female sex has consistently been identified as a risk factor for developing both LC and CFS. Large population-based and cohort studies have reported a higher prevalence of LC among women [69, 70] as well as female predominance in CFS cohorts [65, 66]. This is consistent with the present findings, which showed a higher prevalence of LC–CFS overlap among women than among men.

On the other hand, most previous investigations have included sex as a covariate or reported stratified comparisons, rather than formally testing sex as an effect modifier of clinical severity. In the current study, no significant sex-by-overlap interaction was observed for any of the health status outcomes analyzed ($p > 0.05$). This suggests that although women were more likely to meet criteria for LC–CFS overlap, the magnitude of impairment associated with this phenotype was comparable in women and men. In this sense, sex may influence susceptibility to developing the overlap condition, but not necessarily the severity of its impact once established.

All these findings align with the hypothesis that the pathophysiological mechanisms underlying the LC–CFS phenotype—such as autonomic dysregulation, immune dysfunction, and neuroinflammation—may exert a similarly disabling effect across sexes [63]. Nevertheless, limited statistical power for interaction testing cannot be completely excluded, and further studies specifically designed to assess sex-based effect modification are warranted.

Implications for healthcare systems

The consistent association between LC–CFS overlap and poorer outcomes across multiple health status questionnaires underscores the clinical and organizational relevance of this subgroup for health systems and primary care. These findings are consistent with previous evidence showing that LC is associated

with substantial primary care costs and increased consultation burden, particularly among women and patients with comorbidities [5]. Moreover, it supports the need for multidisciplinary care models coordinated by primary care teams, integrating medical follow-up, rehabilitation, mental health care, and social support. Health systems should consider incorporating structured screening for CFS-like features into LC care pathways, particularly in primary care, to facilitate early identification of patients with greater clinical complexity and higher care needs. Resource allocation strategies should account for the potentially prolonged care trajectories, functional impairment, and increased demand for coordinated care in this subgroup.

Strengths and Limitations

This is a cross-sectional study. Therefore, due to the nature of the study, it is not possible to establish a causal relationship.

PEM was not directly assessed using PEM-specific instruments. Instead, two items from the MFIS reflecting exertional intolerance (Item 6: “I have had to moderate my pace in physical activities” and item 10: “I have trouble maintaining physical effort for long periods of time”) were employed in order to name the need to reduce physical activity and the inability to sustain effort over prolonged periods. Although these items do not explicitly assess whether rest alleviates symptoms, higher scores indicate persistent physical discomfort and may therefore serve as a proxy for fatigue that is not substantially relieved by rest. The selection of these items was based on their content relevance within the MFIS and their conceptual alignment with exertion-related symptom persistence described in CFS. Nevertheless, this approach does not fully capture the multidimensional nature of PEM, and future studies should incorporate PEM-specific instruments to more accurately assess symptom exacerbation and recovery following exertion. Despite this limitation, the consistency of worse outcomes across multiple independent domains—including functional status (PCFS), physical functioning and HRQoL (SF-36), perceived dyspnea (mMRC) and pain level (SCPGS)—supports the identification of a clinically meaningful high-burden phenotype within LC compatible with CFS.

On the other hand, the present study is the first, to the best of the authors' knowledge, to undertake a concurrent analysis of the health status of patients diagnosed with both LC and CFS, encompassing such a large number of multidomain health status questionnaires including HRQoL, functionality, depression, anxiety, pain, dyspnea and nutrition, as well as taking sex difference into account. This provides a more holistic view of the impact on patients with a dual diagnosis compared to those with LC alone.

Future studios should aim to further characterize the LC subgroup with overlapping CFS using longitudinal designs and standardized, condition-specific instruments, particularly for PEM symptoms. Prospective studies would help elucidate the temporal evolution of functional impairment, quality of life, and symptom burden in this high-impact phenotype, as well as identify predictors of persistence or recovery. The integration of objective functional assessments, alongside patient-reported outcomes, may improve phenotypic stratification and inform tailored intervention strategies. Moreover, the

consistent association between LC and CFS overlap and poorer outcomes across multiple health domains supports the need for early identification, risk stratification, and multidisciplinary care pathways. Furthermore, they may assist in the development of targeted rehabilitation and social support strategies in order to mitigate long-term disability and improve quality of life in affected populations.

CONCLUSION

More than half of individuals with LC met criteria for LC–CFS overlap, with a higher prevalence among women. This subgroup exhibited substantially poorer health status, quality of life, and functional outcomes compared with patients without overlap. No sex difference was found when sex-overlap interaction was analyzed. Recognizing LC–CFS overlap in primary care may help identify patients at higher risk of persistent disability and support more targeted management and rehabilitation strategies. Further prospective studies are needed to clarify temporal trajectory of HRQoL and functional impairment and to identify predictors of persistence and recovery, as well potential sex-related differences.

Abbreviations

LC	Long COVID
CFS	Chronic Fatigue Syndrome
HRQoL	Health-related quality of life
APISAL	Salamanca Primary Care Research Unit
WHO	World Health Organization
MetS	Metabolic Syndrome
NCEP ATP III	National Cholesterol Education Program Adult Treatment Panel III
HDL	High Density Lipoprotein
COPD	Chronic Obstructive Pulmonary Disease
SF-36	Short Form Health Survey
PCS	Physical Component Score
MCS	Mental Component Score
HTS	Health Transition Score
mMRC	Modified Dyspnea Scale

MFIS	Modified Fatigue Impact Scale
PSQI	Pittsburgh Sleep Quality Index
SCPGS	Spanish Chronic Pain Grade Scale
GADS	Goldberg's Anxiety and Depression Scale
MUST	Malnutrition Universal Screening Tool
PCFS	Post COVID-19 Functional Status Scale
PEM	Post-exertional malaise

Declarations

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Author contributions

ANC and LGS designed the study. ANC, NSM, SAR, ADM and ABCR collected the data. CLS and SGS supervised the database information. LGS and ANC drafted the manuscript. MAGM and ENM critically revised the manuscript. All authors approved the final version of the manuscript.

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Data availability

The data supporting the findings of this study are available on ZENODO at: https://zenodo.org/records/14282873, DOI: 10.5281/zenodo.14282873.

Ethics approval and consent to participate

The study was approved on 27 June 2022 (CEIm reference code: Ref. PI 2022 06 1048) by the “Ethics Committee for Research with medicines in the Salamanca Health Area”. The standards of good practice in observational studies established by the Declaration of Helsinki and by WHO were followed throughout the study. The confidentiality of participants was always guaranteed in accordance with Organic Law 3/2018 European Regulation 2016/679 and Council Directive 27/04/2016 on Data Protection. All participants signed an informed consent form before being included in the study, after receiving information about the procedures that would be performed on them.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Figures

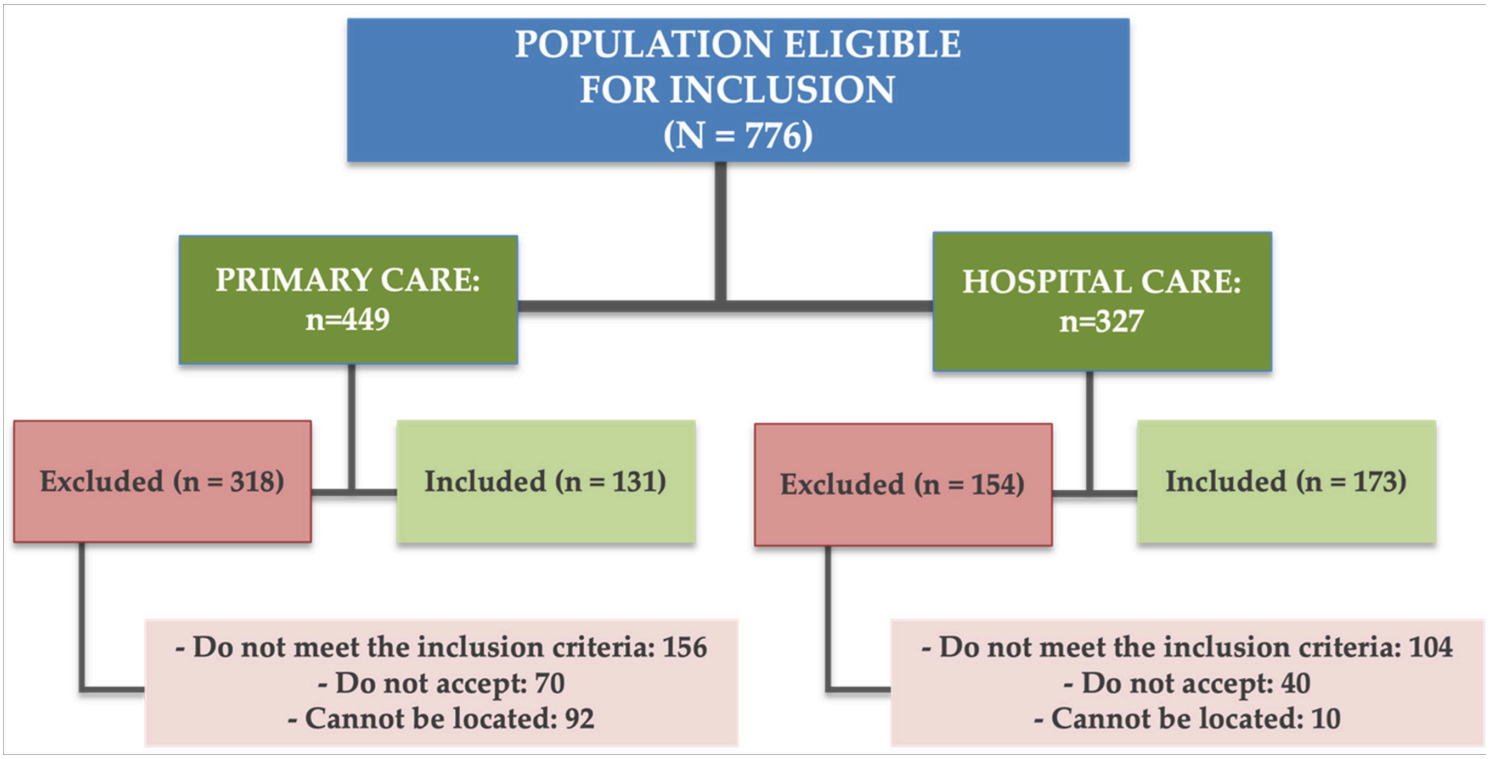


Figure 1

Flow diagram of the participant recruitment process

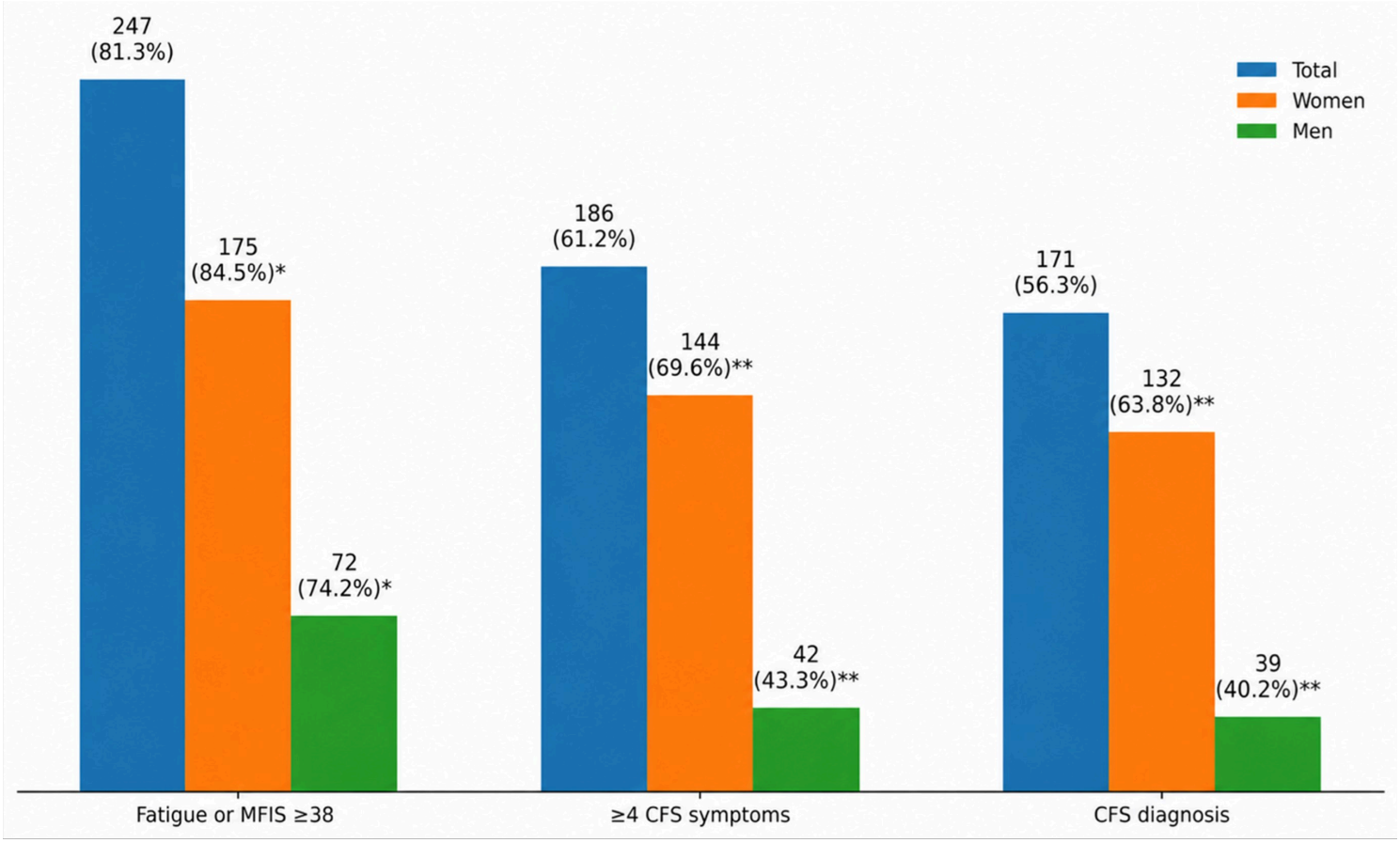


Figure 2

Prevalence of CFS diagnostic criteria and CFS diagnosis among patients with long COVID.

MFIS: Modified Fatigue Impact Scale; CFS: Chronic Fatigue Syndrome. *p* value: differences between Men and Women; *p <0.05; **p <0.001 for sex differences.

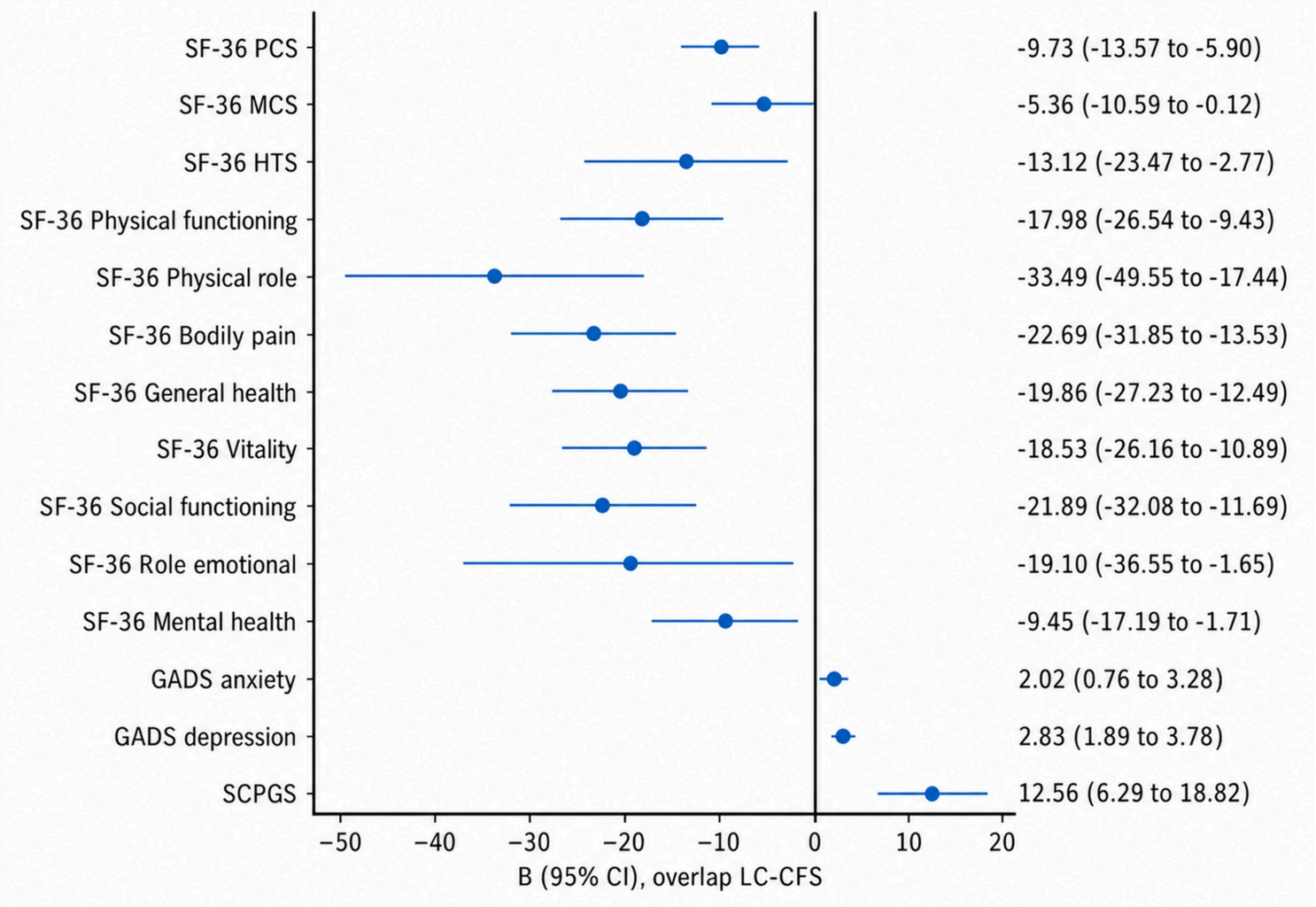


Figure 3

Multiple linear adjusted regression model assessing the association between LC–CFS overlap and health status questionnaires

LC: Long COVID; CFS: Chronic Fatigue Syndrome; SF-36: Short Form-36 Health Survey; PCS: Physical Component Summary; MCS: Mental Component Summary; HTS: health transition score; GADS: Goldberg Anxiety and Depression Scale; SCPGQ: Spanish Chronic Pain Grade Questionnaire.

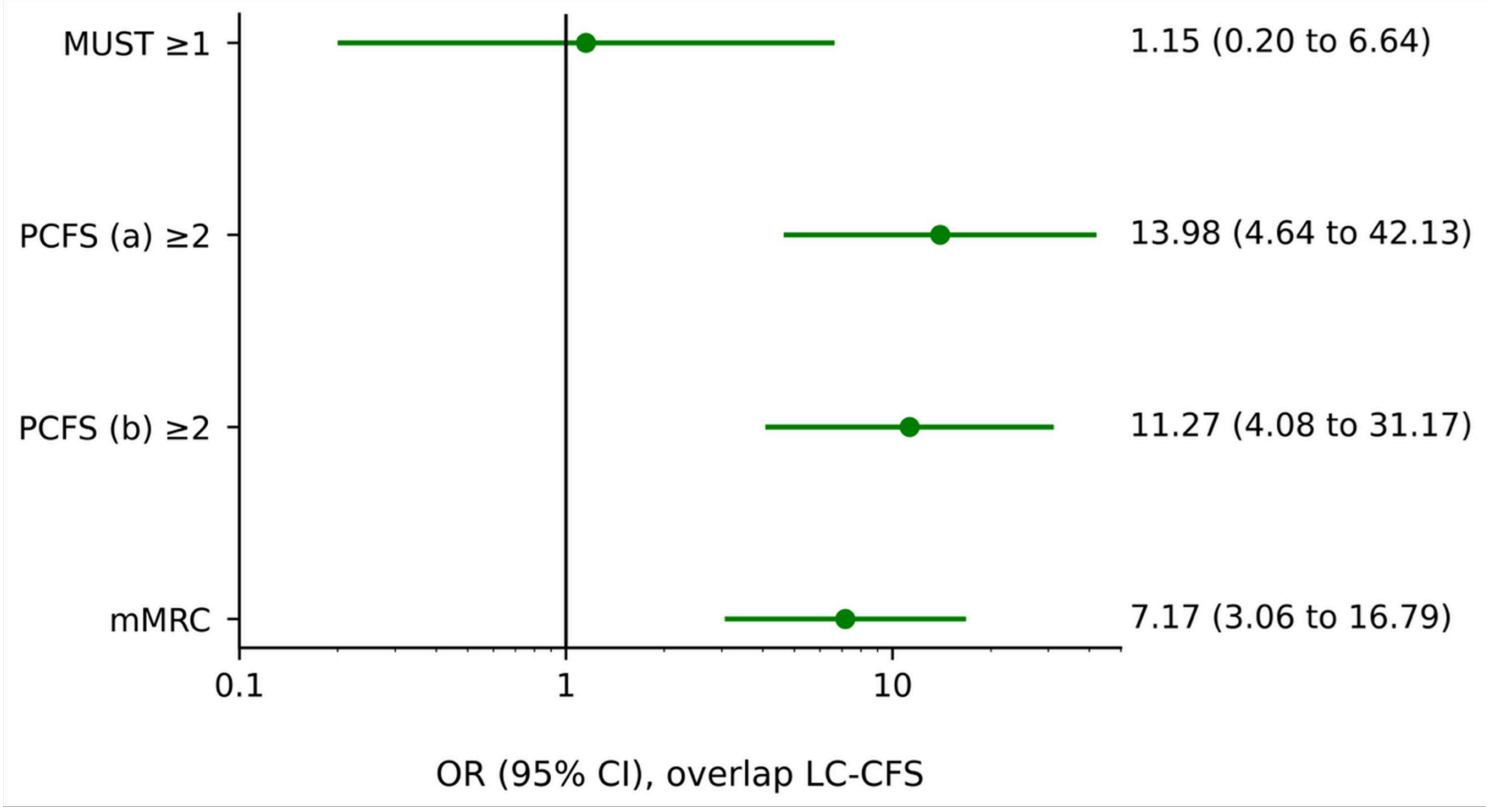


Figure 4

Binary and ordinary adjusted regression models assessing the association between LC–CFS overlap and health status questionnaires.

LC: Long COVID; CFS: Chronic Fatigue Syndrome; PCFS: Post-COVID Functional Status Scale; (a): a component; (b): b component; MUST: Malnutrition Universal Screening Tool; mMRC: modified Medical Research Council Scale.

Supplementary Files

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